

Translated

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Lowest Common Ancestor - Farach-Colton and Bender Algorithm

Let G be a tree. For every query of the form (u, v) we want to find the lowest common ancestor of the nodes u and v , i.e. we want to find a node w that lies on the path from u to the root node, that lies on the path from v to the root node, and if there are multiple nodes we pick the one that is farthest away from the root node. In other words the desired node w is the lowest ancestor of u and v . In particular if u is an ancestor of v , then u is their lowest common ancestor.

The algorithm which will be described in this article was developed by Farach-Colton and Bender. It is asymptotically optimal.

Algorithm

We use the classical reduction of the LCA problem to the RMQ problem. We traverse all nodes of the tree with DFS and keep an array with all visited nodes and the heights of these nodes. The LCA of two nodes u and v is the node between the occurrences of u and v in the tour, that has the smallest height.

In the following picture you can see a possible Euler-Tour of a graph and in the list below you can see the visited nodes and their heights.